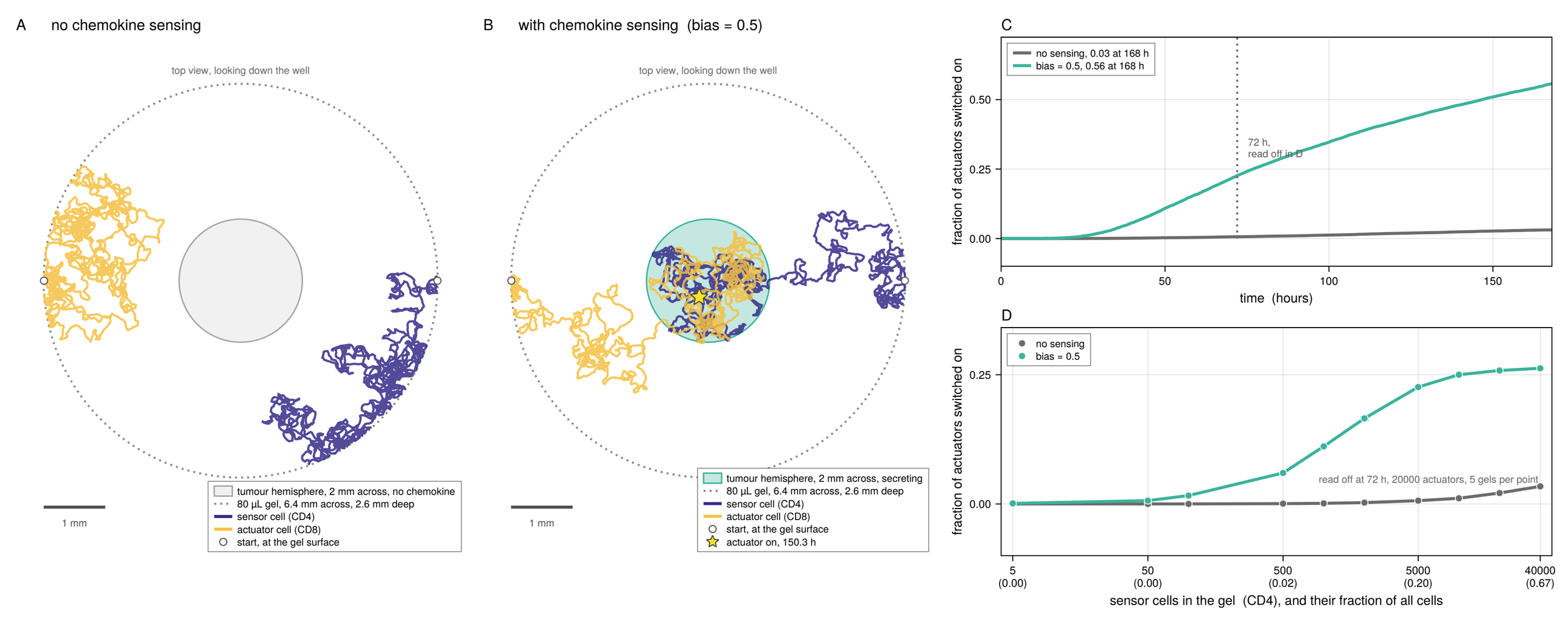

A
no chemokine sensing
top view, looking down the well
1 mm
tumour hemisphere, 2 mm across, no chemokine
80 µL gel, 6.4 mm across, 2.6 mm deep
sensor cell (CD4)
actuator cell (CD8)
start, at the gel surface
B
with chemokine sensing (bias = 0.5)
top view, looking down the well
1 mm
tumour hemisphere, 2 mm across, secreting
80 µL gel, 6.4 mm across, 2.6 mm deep
sensor cell (CD4)
actuator cell (CD8)
start, at the gel surface
actuator on, 150.3 h
C
no sensing, 0.03 at 168 h
bias = 0.5, 0.56 at 168 h
fraction of actuators switched on
0.00
0.25
0.50
72 h,
read off in D
0
50
100
150
time (hours)
D
no sensing
bias = 0.5
fraction of actuators switched on
0.00
0.25
read off at 72 h, 20000 actuators, 5 gels per point
5
(0.00)
50
(0.00)
500
(0.02)
5000
(0.20)
40000
(0.67)
sensor cells in the gel (CD4), and their fraction of all cells